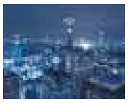
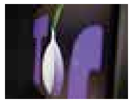


ONPC Wi-Fi boost
Researchers have developed a new wireless protocol that can extend the reach of Wi-Fi devices. **https://dgit.in/nov19-13**



BBC fight censorship
BBC fights censorship by launching a news site on the dark web, which you can access via Tor. **https://dgit.in/nov19-14**

down from the top-right on iPhone X or newer iPhones or swiping up from the bottom on iPhone 8 or older iPhones and then activate Live Listen.

Xbox Adaptive Controller

Microsoft recently came out with their Xbox Adaptive Controller for those living with mobility impairments. The controller can be paired with an array of external assistive devices such as switches, buttons, mounts and joysticks which can control the trigger inputs on the controller. The company has also filed a patent application for a Braille-input and output controller for the visually-impaired.

Bionic Exoskeletons

A company named Ekso Bionics develop bionic exoskeletons to help victims of stroke, brain and spinal cord injury get back on their feet. Patients secure their limbs to a bionic suit and the suit utilises weight sensors to perceive when the user shifts his or her weight forward and then triggers the device to step ahead. It has been used extensively in injury rehabilitation centres.

LUKE prosthetic arm

Inspired by the scene in *The Empire Strikes Back* when Luke Skywalker's bionic hand responds to the probes of a medical robot, researchers have created the LUKE, which restores the sensation of touch! Researchers from the University of Utah utilised a DARPA-funded prosthetic arm, LUKE, and have figured out a way to allow amputees to feel around 100 sensations. A device is implanted in the amputee's residual nerves and electrodes and placed in muscles which enables an information loop that transfers signals to the brain which are recognised as sensations of touch.

NeoMano Robot Glove

This is a robotic glove that enables people to grip things more firmly and pick them up and could be especially

useful for people with multiple sclerosis, cerebral palsy, spinal cord injuries or those recovering from a stroke. The glove fits over the user's index, middle finger and thumb and a single press on a Bluetooth controller will activate the titanium wires housed inside the glove which closes the wearer's hand. Users with mobility problems can grip items, twist open bottles and containers, turn pages, open doors and much more.

Dot

Dot is a motorised Braille smartwatch for visually-impaired users. It comes with a rechargeable battery, a touch sensor, a vibration motor and Bluetooth. Under the dial, there are four motorised modules present, with six possible dots each. Each dot can be raised as well as lowered individually

and the Dot is capable of displaying up to four braille characters at once. The wearer can read the dial as they would a piece of paper with braille. Upon connecting it to your smartphone, users will receive Braille notifications from their smartphone on the watch.

iBot

Created by innovator Dean Kamen, the iBot is a self-balancing, stair-climbing wheelchair made for the physically-challenged. This device tackles one of the most trying situations for people with limited mobility – stairs. They are nearly everywhere and navigating them on a traditional wheelchair can seem close to impossible. The iBot utilises self-balancing technology which is similar to that found in a Segway and aims to change the way wheelchair-bound people navigate stairs.

ROOM FOR GROWTH

Technology for the disabled and assistive technology has taken major strides over the last few years. Several tech giants have even joined forces to address accessibility. For example, Facebook, Google, Microsoft, Adobe and Oath together launched an accessibility program in 2018 as part of TeachAccess. However, there is always room for improvement and this area of assistive technology is no different. Nutsiri "Earth" Kidkul, lead tech instructor at the Braille Institute of Los Angeles Centre told CNET that mainstream tech companies could do a better job of including accessibility features in software updates. She also stated that sighted people usually get new features way before blind or visually-challenged individuals can access them.

It is imperative that companies creating technological innovations and products keep accessibility for the disabled at the forefront while developing gadgets and services. Companies such as IBM and Microsoft have hired 'chief accessibility officers', a relatively new occupation, in recent years and most tech giants have a sizeable accessibility team. Smaller companies must attempt to follow suit and employ a few, or at least one, disabled individual to understand their perspective more holistically. Also, they must be hired for all roles, and not just ones tied to accessibility since a product or service's accessibility quotient isn't just determined by the efforts of one team but by an amalgamation of teams working together that are all collectively responsible for the creation of said product or service.